

Institute of Actuaries of India

Subject SA1 – Health and Care Insurance

October 2015 Examinations

INDICATIVE SOLUTIONS

Solution 1:

i)

Target segment & selling process:

Discuss with Sales team and try to have a fair understanding regarding target segment. At the same time, need to assess how the product will be sold and pitched to customers (e.g. whether there will be a special distribution channel for this product). It is very important to understand why offering the Global coverage will be a good marketing USP, considering the fact that Travel Insurance products are easily available in the market and inclusion of Global coverage in the product will make the product expensive.

Potential conflict with Travel Insurance product:

If the Company is already active in the Travel Insurance business, needs to investigate whether the introduction of the new product will affect the sale of Travel Insurance product, hence profit margin.

Comparison with premium of travel insurance product will be an important consideration here.

Competition:

Do a market research to assess whether any of the competitors is already offering such coverage in the market. If so, a proper feature and pricing comparison will be required to be done to ensure that product doesn't lose its USP in the market, when launched.

Coverage scope & liability:

It is very important to design the right coverage of this component e.g. whether the Global component will have both in-patient (IPD) and out-patient (OPD) coverage. At the same time, Company needs to decide whether the Global coverage will have other common coverage traditionally offered by Travel Insurance product eg. Baggage/passport loss, Flight delay etc.

As the coverage will be outside of India, potential liability could be quite high as against domestic coverage's liability. Hence, need to be careful while deciding on the ultimate liability of the insurer under Global coverage.

PED & geographical scope:

One important aspect here is whether Company decides to cover PED conditions from day one or not. While purchasing this product, this may be one of the primary needs of customers.

Company needs to also decide on the Geographical scope of the coverage.

Underwriting condition:

As the product will have Global coverage, it is very important to understand the pre-acceptance of underwriting conditions. Need to decide on the age/sum insured limit, which will trigger Medical underwriting. If the Global coverage is offered as an optional cover to the domestic coverage, then it is very important to understand whether the optional coverage will be offered throughout the year or only at the time of purchase of first policy and renewal.

Claim administration:

Given the coverage outside of India, Company needs to establish a proper claim adjudication process through an International TPA. Company needs to provide adequate cashless networking hospital facility across the Globe (as per the coverage scope) for its customers who plan to opt for Cashless facility.

Company may be required to tie-up with some Re-pricing Agencies present outside of India, who may help it to negotiate with hospitals while settling the bill.

Reinsurance coverage:

Given the potential high severity of claim, it will be very important to get right Reinsurance Coverage for this component. At the early stage of the product, Company may want to consider both proportional and non-proportional Reinsurance coverage, which will help it to increase its underwriting capacity, spread the risk and avoid large claim which may impact Company's financials. Company's Capital position will also be an important determining factor.

Capital consideration:

If the potential volume is very large, it becomes important to assess the capital requirement, especially the economic capital requirement if actual experience turns out to be worse than expected. Need to consider the solvency position of the company in the worst case scenario.

Data & pricing related assumptions:

Need to consider whether company has adequate experience outside of India. One important pricing assumption is the proportion of customers who will actually travel outside of India and where they will travel to.

In absence of adequate data and experience, help from Consultant /Re-insurer may be sought for.

Some other factors like Regulatory approval, Administration system, availability of subject matter experts, whether Global or specific country coverage, Domestic currency stability level can be considered as well for answering this question.

[5]

ii)

- Availability of data would be a big challenge during the pricing exercise. If the Company does not have any overseas experience, Company needs to make several pricing assumptions without data support. In absence of data, this proposition will be exposed to high uncertainty.
- Potential Claim liability could be very high as the liabilities will be denominated majorly in USD and Euros. Absence of adequate Reinsurance may affect product's financial viability in a major manner especially during the initial phase of the product, if multiple large claims happen at the same time.
- Needs to recognize synergy with other products. It needs to be seen if the product may affect the sale of existing health or travel insurance product. Company will be required to ensure that target segment for the travel insurance product remains different through different product proposition and probably through a different distribution channel approach.
- Given coverage outside of India, claim administration will be a challenge specially building up a cashless network outside of India. Company may be required to deal with multiple partners due to providing coverage in multiple regions. Language may be a challenge in certain regions.
- Currency Risk:
 - a) Exchange rate assumptions in pricing may be different from Actual
 - b) Company needs to invest in assets denominated in other currencies to match its liabilities. Existing investment regulation may not allow the Company to do so.
- Presence of Global coverage will make the product expensive which may not be affordable by certain segment. Need to ensure that product pricing is not out of line while compared to similar product(s) in the market.

[6]

iii)

- If internal data is not available, take a look at external data in the form of published statistics & Competitor's (who have alike product) premium rate. If nothing is available, help from consultants/Reinsurers may be sought for. Also refer to Medical journals, Government statistics in respective countries to assess the impact of hospitalization cost.
- Needs to ensure that that product USP is different from existing Health Insurance or Travel Insurance product. If the Global coverage is offered as an

optional cover, it may be offered at point of sale or at renewal only and through this approach Company will be able to keep the target segment different from that of Travel Insurance product. If the Global coverage is offered as an optional cover, it also addresses the issue of affordability.

- Given the uncertainty level regarding claim experience outside of India, following product restrictions could be put across:
 - a) Don't offer very high sum insured coverage for outside of India
 - b) Ensure that adequate Reinsurance is available (XOL arrangement with multiple re-instatement options)
 - c) Probably a co-pay/deductible can be applied on per claim
 - d) Maximum number of days during a single trip can be limited
 - e) Underwriting approach can be made stricter for this product
- Tie up with reputed & established TPAs/Re-pricing agencies. Establish a dedicated claim team for the product with specialists in the field.
- Monitor the product experience at periodic intervals. If the experience worsens (Eg Currency depreciates), needs to assess the phase duration during which INR Currency will remain weak and if a long period is envisaged, product may be re-filed for pricing revision.

[6]

iv)

(a)

- Generally Pre-existing disease (PED) conditions are covered from day one in emergency cases in the case of travel insurance product and there is a practice of putting a sub-limit in such cases (e.g. 10% of medical coverage).

In the proposed product, PED conditions may be subject to a typical initial waiting period restriction as observed in standard hospitalization product. However, once the waiting period is over, PED conditions are expected to be covered fully which means emergency and non-emergency situations and 100% sum insured coverage.

- Travel Insurance product is a short term product where the term of the product is aligned with trip duration and generally it is not renewed. As against that, proposed product is expected to be yearly renewable contract and lifetime renewable.

- Some of the standard travel insurance product features (e.g. Trip cancellation/flight delay/Passport loss/Baggage loss) may not be part of the proposed product.
- Travel Insurance product will provide region specific coverage (depending on the plan chosen for) whereas the proposed International coverage is expected to provide Global coverage.

[2]

(b)

Trip duration is an important rating factor in travel insurance product pricing framework. Premium increases with increased duration as the risk exposure increases.

As the proposed product is expected to be a yearly renewable contract, trip duration will not be considered as a factor in the premium rating structure.

In travel insurance product, Region (Destination) is an important rating factor. However, the proposed Global product is not expected to have Region (destination) as a rating factor. It is expected that there will be default pricing assumption regarding Region mix.

[2]

v)**(a)**

- Starting point will be Company's internal experience with respect to hospitalization product(s) providing domestic coverage. Given the fact that Company has been into business for more than eight years, Company is expected to have credible experience as far as domestic coverage is concerned.

Claim severity:

Adjust the past data to take care of the following:

- Ensure that any one-off event (e.g. large claim) is taken care of appropriately. They

may be excluded for the analysis purpose or averaged out over the years.

- Claim experience is to be adjusted to remove the impact of product feature restrictions e.g. (room rent cap/deductible/co-payment/sum insured) which will ensure that data represents actual claimed amount
- claims reported recently should be adjusted for IBNER, if any
- Past experience data needs to be adjusted by past medical inflation (as experienced) to ensure that past data represents claim experience as on analysis date. Needs to make assumption on when the product will be

launched. Accordingly project the claim data to future claim inception date (six months after the average policy inception date) by making assumptions regarding future medical inflation. Discuss with Claims and hospital networking functions to assess future medical inflation.

- Analyse the data year wise and through all the important rating factors (e.g. age, sum insured, customer's residence location) which are expected to make an impact on claim severity experience
- Adjust the amount by any other product design restriction like co-payment/deductible/room rent cap etc.
- Final severity estimate which would vary by age/location should be adjusted further for sum insured to calculate the net impact due to sum insured capping as the starting point was actual claimed amount.

Claim Frequency:

- Adjust the past data by deductible (if any) to ensure that none of the claims gets unnoticed due to deductible
- Analyse the experience by rating factors like age, location, sum insured and accordingly calculate the exposure for different time cohort
- Corresponding claim count should be adjusted for IBNR
- Accordingly Claim frequency gets calculated for each time period and corresponding rating factor like Age
- Final claim frequency estimate should be adjusted for any trend observed over the entire time period

The claim severity and claim frequency estimates will be used to derive the risk premium before any additional adjustment.

[4]

(b)

Select lives due to initial underwriting: Claim experience of the insured is expected to be better during the initial policy duration due to pre-acceptance underwriting process. Hence, claim experience within the same age band will vary by policy duration.

Initial waiting period restriction: There is a standard waiting period restriction for PED conditions and time bound exclusion for named ailments like Cataract, Cyst. Presence of the waiting period also leads to difference in claim experience by policy duration within the same age band.

Selective withdrawal/selective lapsing: Over time, good risks tend to leave the portfolio and poor risks stick to the portfolio which worsens the claim experience

with increase in policy duration. Increase in premium rate accelerates the worsening rate.

Hence, there will always be a difference in claim experience by policy duration within the same rating factor (e.g. Age). Policy duration is not used as a factor in the premium rating structure. Hence, it is important to assess the impact of these factors in the claim experience of the existing products and it needs to be decided to what extent the impact needs to be recognized in pricing assumptions of the proposed product. It is to be kept in mind that a higher margin in pricing assumptions to account for these factors may increase the premium of the proposed product to a level which may make the entire proposition quite unattractive. Hence, a balancing approach is required here.

[4]

(c)

- Need to assess the treatment cost in different regions outside of India (e.g. USA/Canada, Europe, Africa, Asia etc). Refer to Published statistics to determine estimate of the treatment cost of top 10-15 ailments/surgeries outside of India. Help from Reinsurers/consultants/international TPA's (with whom contract will take place) may be sought for to assess the same.
If the Company has a foreign partner, help can be taken from them as well. In addition, experience of Travel Insurance product (if any) should be referred to. Accordingly, derive cost multiplier estimate for different regions as compared to treatment cost in India.
- Consider the adjusted unclaimed claim severity of existing product(s) in India, as has been done for pricing of the domestic coverage. Claimed amount records in India need to be adjusted by specific currency exchange rate (e.g. INR to Dollar, INR to Euro etc). Secondly, cost multiplier needs to be applied. Accordingly, claim severity for different rating factors (& combinations) is estimated for specific geographical region.

Other adjustments

- Needs to make an assumption regarding the proportion that will use the Global coverage. Above proportion may be split into two components:
 - i) Travel segment ii) planned surgeries
- Travel Segment may be perceived to be healthier; hence an appropriate discount can be applied both on the claim frequency and claim severity assumption. Travel insurance product's Claim experience may be referred here in order to ascertain the discount factor. Similarly, additional loading may be applied for the segment which would opt for planned surgeries. It is

expected that this segment will undergo major surgeries like Bypass/Cancer treatment.

- Another important assumption needs to be made here regarding the duration outside of India for the proportion which are expected to claim outside of India. Accordingly, a proportionate reduction factor is required to be applied on Domestic coverage risk premium to reflect the fact that the insured will be outside of India for that assumed duration and he/she is charged for that duration under Global coverage.
- In addition, if the premium rate of the proposed rate doesn't vary by geographical region outside of India, weight needs to be assigned against different regions (proportion of people going to respective region) and accordingly, weighted average risk premium will be derived to represent the Global coverage.

[6]

[35 Marks]

Solution 2:

i)

- Simplicity: Given the general level of financial illiteracy level, it is important to make the overall process very simple with respect to product benefits, sales process, operations and claims. It will instil confidence in the mind of people which will ensure a large take up rate. This is important for the long term sustainability of the proposition.
- Large enrolment: The very purpose is to cover society at large. Also it is important as there may not be many safeguards related to gatekeeping (eg. No PED waiting period), so that, big enough pool is generated. A large enrolment is a very important factor to ensure financial viability of the proposition despite of low premium and absence of any form of pre-acceptance underwriting.
- Risk Equalisation: The very purpose of this product is to cover health insurance needs of the population on a sustainable basis, hence, there has to be some pooling mechanism with relation to different insurers across the years.
- Provider Contracting: Important to have some structure like package rates as in case of CGHS or RSBY. Large volume will ensure that these package rates are kept on the lower end and at the same time sustainable. This will ensure a lower premium on a long term basis as affordability to pay premium is a key factor for the success of this product.

- **Use of Technology:** This is going to be a vast scale project across India. Data management and sharing will pose a great challenge. It is very important to leverage on IT technology in all the facts including sales, claims payment, operations, fraud management, etc.

Facilitation in promotion of distribution channel (e.g. Bancassurance model) was also given due credit.

[10]

ii)

Premium has to be affordable and at the same time sustainable which warrants some level of gatekeeping. Large volume and social consideration mean minimal underwriting. Therefore, it becomes important to have some waiting period for coverage of pre-existing. It is also important to know who is paying the premium because that may have direct impact on level of awareness and thus utilisation. It will be difficult to relax underwriting and waive off waiting period if the premium is directly paid by the insured. The underwriting here doesn't mean medical test, a declaration of good health may be enough.

Risk pooling will ensure that insurers are not very conservative in determining premium in the beginning. Also there should not be any differentiation with respect to product features, premium, etc. among insurers.

[5]

iii)

- **Awareness:** With large resources at its disposal, Govt. can do a much better job to bring awareness as compared to insurers and this will ensure large volume.
- **Bringing stakeholders together:** There may be different stakeholders with conflicting interests, like, insurers, banks (as sales channel), health service providers, IT service companies, TPAs etc. It is important for the success of this scheme that they all are aligned and work together to make this scheme a success. Govt. can play a very important role here.
- **Risk Pooling:** Govt. can act as a reinsurer here for example, Stop Loss Coverage can be provided by the Govt. Also there can be a limit on how much profit an insurer can make in any particular year beyond which they need to contribute to pool. This will ensure greater transparency and long term sustainability.

- **Ensuring different segments of society benefit:** Govt. may subsidise premium partially or fully for different segments, like BPL, Senior Citizen, etc. to make sure large participation.

- **IT Enablement:** Given the uncertainty around the volume and performance of the scheme in terms of profitability, insurers may not be willing to commit large sum for bringing IT infrastructure initially. Given the fact that IT is pre-requisition for scheme's success, Govt. can take over this task at least initially. An overarching structure will also help in curbing frauds.

[10]

[25 Marks]

Solution 3:

i)

Branding is important:

- To increase market reach
- To influence and educate customers
- To gain competitive edge
- To have long term association with client
- To promote alternative distribution channels (Direct Marketing/Online and Tel-Marketing)
- To reduce expenses (primarily commissions, etc.)

Following points should be considered by a company while developing a product in line with its brand strategy:

- Company needs to understand its target customer segments and their choices/decisions clearly.
- Creating value proposition (in the form of insurance products and services) around these customer segments.
- Differentiation in the brand strategy from the competitors so as to develop its unique market for its products
- Understanding existing capabilities(in the form of products and services) and areas to focus/improve
- Establishing brand guidelines for all customer touch-points and consistently applying them throughout the product offering cycle
- Impact on industry relations and understanding underlying risks

[6]

ii)

- Mis-selling: Product features are covered inappropriately or insufficiently to misguide customers
- Mis-guiding Regulator: Approved health insurance product may not explicitly cover the insurance benefits (like drug offering could not be part of insurance product) which are promoted or offered with partners under existing health insurance regulations
- Non-alignment with co-partners in terms of business priorities impacting customer service delivery and hence company's brand image and positioning
- Tie-up with dis-reputed partners impacting company's reputation
- Lapse in customer service due to over-dependency on partners' capabilities (like partner's distribution network which is only active in Tier 1 cities, etc.)
- High marketing expenses resulting in expense over-run
- The target customer segment not tapped appropriately by marketing and branding campaigns resulting in lower than expected business volumes
- Other Regulatory lapses at company's end: Not adhering to regulatory guidelines in the context of branding resulting in penalties and fines.
- Regulatory lapses at partner's end; e.g. selling of drugs under trial or overcharging for drugs as compared to real MRP to overstate availed drugs benefits under insurance policy.
- Unwarranted customer experiences: Lapses in customer service like delay in timely drug delivery resulting in death of customer

[5]**iii)****a)**

Digital marketing involves use of digital technology like internet (covering company's website or blog, search engines, emails, etc.), smart phones, web applications and social media to advertise and promote products.

[1]**b)**

Advantages:

- Better understanding of evolving needs and preferences of valued customer segments: This results in offering long term value proposition to the existing and potential customers resulting in strong customer association
- Enhanced ability to deliver strong and compelling marketing messages: Which touch points are most effective? Where and how to invest?

- Enhanced ability to filter and focus on good risks segments and influence their behaviour
- Utilizing multiple digital distribution channels to reach wider customer base
 - o Promotes customer retention
- Ease of controlling technology up-gradation in line with customers' preferences
- Expense efficiency
- Able to establish a differentiating brand for itself.

Dis-advantages:

- Complex product structure may limit complete understanding of underlying benefits when sold via digital platform
- Primarily taps young and sophisticated customer segments who can select against company(due to better understanding)
- High technology spend to keep pace with technology up gradation
- Challenge to balance new business and retention strategy due to varied needs of underlying customers
- Competition can easily duplicate digital marketing strategy due to its openness and transparency. Hence continuous efforts are needed to stay ahead of the market.
- Data privacy issues resulting from fetching and storing customers personal information online
- Engaging senior staff in the organization to understand and drive digital marketing campaigns could be challenge due to their limited understanding
- Cannibalization of conventional distribution channels(like agency, brokers, etc.) resulting partners grievances impacting business volumes

[6]

iv)

Points to consider:

- As a starting point, they can review how companies like ABC are allocating their total marketing budgets towards digital marketing and decide on an appropriate allocation for the same from total marketing budget.
- How company would be utilizing existing technology to adapt to customers' requirements.
- How much further investment would be needed (for activities like website up-gradation, web-applications build-ups, online advertisements, etc.)? All requirements need to be listed down to understand additional expenses needed.
- Mix of distribution options (paid searches, website/blog, and social media) need to be evaluated and ascertain an efficient expense allocation among them.
- How much budget needs to be set aside for Digital Marketing Head as a senior management resource would be needed to solely drive digital marketing initiative.

- How much savings could be expected from digital marketing activities and how these savings could be utilized?
- How marketing analytics would be done and what investment is needed in this context (in terms of resources and technology)?
- How and what all insurance products would be sold via digital media and how to drive buyers there? Not all insurance products could be marketed at the same time.
- How external agencies and service providers would be utilized where the company lacks experience and expertise (like mobile application development)?
- Impact of this marketing campaign on other marketing campaigns run by company.
- Regulatory and reputational risks involved should be determined and discussed.
- Impact on industry relations and response of competitors.

[6]

v)

Key risks:

- Product would be for customer segments which would be tapped via digital media (primarily youth and financially sophisticated individuals). These customer segments could be more demanding and more customized product offerings would be needed to tap them
- This profile could be significantly different from the existing customer segments covered by the company and could be difficult for the company to ascertain the underlying risk experience
- Product needs to be simple in structure so as to allow appropriate understanding of product features by customers
- Customers' grievances could be high and could be blown out of proportion due to demanding nature of online customers
- High scale IT investment would be needed which may cause initial capital strain
- Limited flexibility to vary premium as the prices would be open to comparison with competitors online
- Product could be easily copied by competitors and constant efforts are needed to innovate to stay ahead in the race
- Cannibalization of other products if this product manages to create high results. This would impact business plan of the company
- Interests of other distribution channel (agency, brokers, etc.) need to be protected so as not to lose them to competition as a result of this strategy
- Reaction of competitors need to be considered
- Regulatory issues with respect to customers' data privacy, online marketing need to be adhered strictly which may limit company's business plan.

[6]

vi)

- Such proposition is expected to have long gestation period. The initial set-up cost will be quite high and the incremental cost is expected to reduce over time with increase in business volume. Hence, a short term approach will make the proposition look quite unattractive. Need to take a long term approach, at least for 5-6 years. A Discounted Cash Flow (DCF) model can be built here
- Prepare a business plan for a time period of minimum of five years. Revenue and expenses will be required to be projected separately in each year. Discussion with all the relevant stakeholders is extremely critical here.
- While estimating revenue, new business volume and renewal business volume needs to be separately projected for each year. It should be kept in mind that persistency ratio for this channel could be different from other conventional channels like Agency. Accordingly, assumption needs to be made.
- While making expense related assumptions, technology set up cost and marketing expenses need to be carefully estimated. Needs to ensure that expenses don't get underestimated. Need to segregate expenses by new and renewal business category.
It is expected that acquisition expenses will reduce with increase in policy duration
While projecting claim experience, needs to consider if this channel will experience a different claim experience compared to other channels
- Net cash-flow needs to be projected for each year separately. It will also consider the incremental capital infusion which will be required to fund the losses along with meeting Regulatory solvency requirement. A terminal value should be estimated at the end of projection period which can be determined by inputs like final year profit, future annual profit growth assumption and risk discount rate estimate.
- Risk discount rate to be applied in the DCF model should be commensurate with the uncertainties regarding future assumptions. Net present value calculated in this manner may be compared with internal yardstick before taking the final decision.

[10]**[40 Marks]**
